

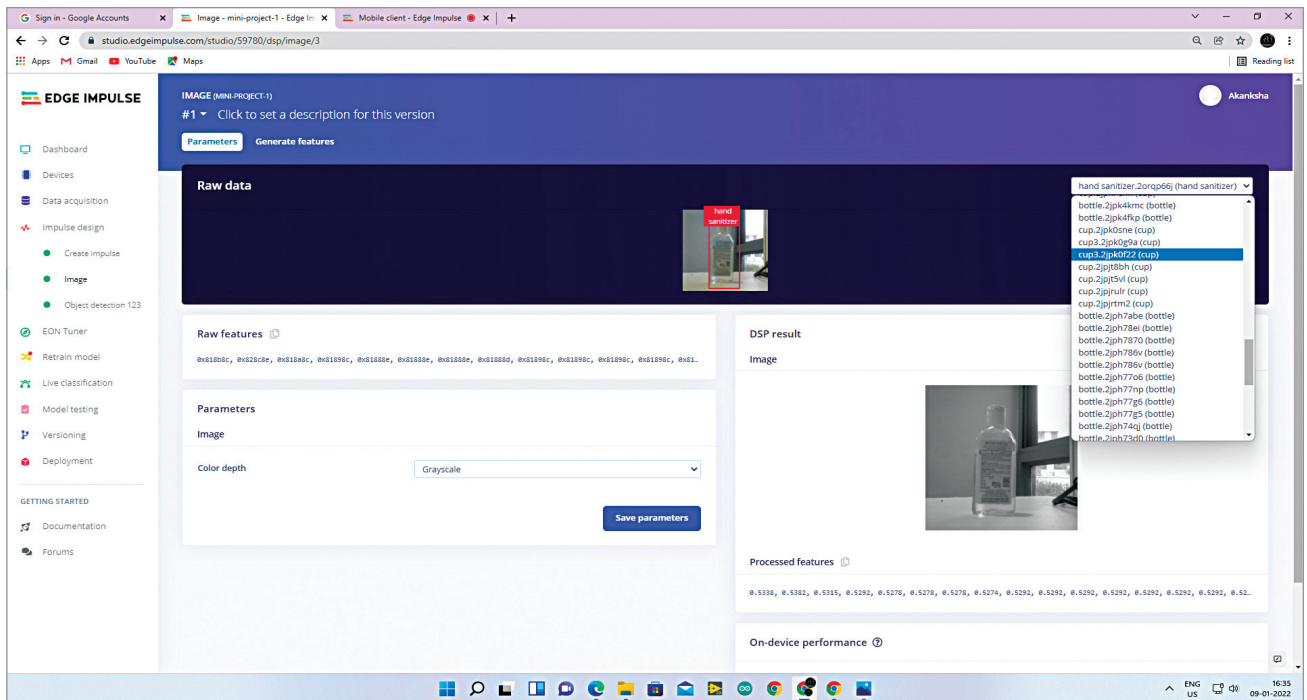


Fig. 5: Generating feature

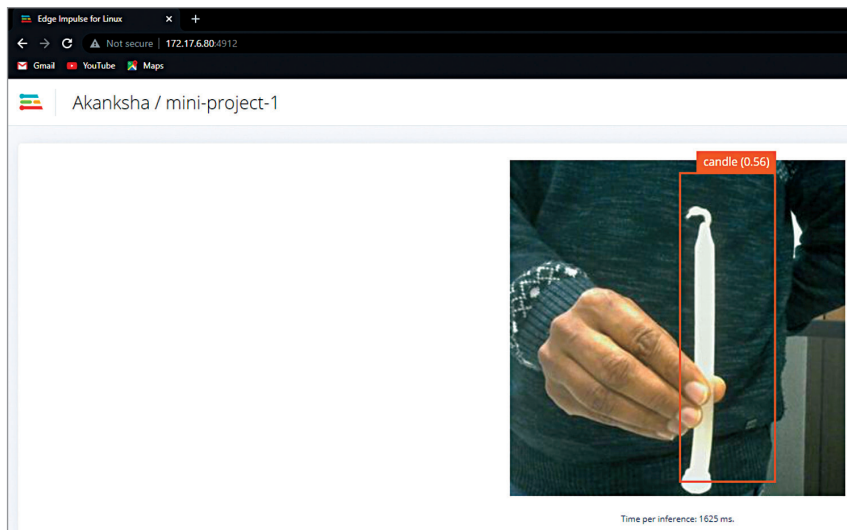


Fig. 6: ML model showing live classification

ject's name and then click on Save Impulse.

In the Image section, configure the processing block and select raw data on top of the screen. Now save the parameters either in RGB or Grayscale.

Due to different dimensions of the images, dimension reduction takes place.

In the Object Detection sec-

tion, the number of training cycles and learning rate can be set. In the author's prototype the number of training cycles was set at 25 and the learning rate at 0.015.

You can now start training the model. After training the model you can get a precision score. For validating your model, go to Model Testing and select Classify All. Then go to live classification.

Put an object in front of the camera and the video of object can be seen in Live Classification section of Edge Impulse without any label of the classified and detected object, such as the bottle, cup, or a person. If you want to see it with IP address, run the command `edge-impulse-linux-runner`

This command builds and downloads the model in Raspberry Pi and gives an IP address to see live classification. When you open this IP address in browser you can see the output `http://192.168.1.19:4912` **EFY**

Akanksha Gupta is M.Tech in Electronics and Communication from NIT Jalandhar. She is a Research Scholar in the Electrical department of IIT Patna. She is interested in designing real-time embedded systems and biomedical signal processing

Sagar Raj is Founder and Director at Shoolin Labs, Jaipur (Rajasthan). He is Founder and Director at Lifegraph Biomedical Instrumentation, Buxar, and IoT Trainer at Lifegraph Academy, Incubation Centre, IIT Patna